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Local topographic and biotic neighborhood factors influence tree survival in a tropical dry forest, western Thailand

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Abstract

Background Understanding variations in tree mortality and survival is essential to interpreting community dynamics and demographic processes. Tree size and both biotic and abiotic factors are among the local-scale drivers proposed to be essential to explain patterns of tree survival. However, detailed studies of their actual importance are lacking, particularly in tropical dry forests.

Methods We investigated the influence of local topographic (i.e., abiotic) and biotic factors on tree survival at the community, guild, and species levels using data from a 4 ha (200 m × 200 m) permanent plot established in a mixed deciduous tropical dry forest at the Mae Klong Watershed Research Station, western Thailand. Censuses were conducted within the plot in December every 2 years from 1994 to 2022, in which tree mortality was assessed and measurements were performed on all trees with diameter at breast height ≥ 2 cm. Factors influencing tree survival were analyzed using a generalized linear mixed model.

Results We found that tree size was the most important factor influencing tree survival, with a significant positive relationship for all but the largest trees. Among abiotic factors, only elevation had a significant positive effect on the longevity of shade-tolerant species. Of the biotic factors examined, tree density and total basal area significantly impacted tree survival in certain guilds and/or species.

Conclusions Overall, the relative significance of factors influencing tree survival patterns in these tropical dry forests differed significantly across species guilds, tree size classes, and shade tolerance groups. These findings inform forest management in the study region and offer essential data for further research on forest dynamics.

Keywords Abiotic factor, Biotic factor, Forest dynamics, Long-term monitoring, Tree demographics, Tropical forest

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Introduction

Tropical dry forests (TDFs) are typically found in the seasonal tropics and are characterized by prolonged periods of severe drought and low annual rainfalls, often < 100 mm (Sánchez-Azofeifa et al. 2005). These forests cover more of the tropics and subtropics than do wet rainforests (Miles et al. 2006) and are frequently impacted by fires and human activities (Mueller-Dombois and Goldammer 1990). In TDFs, changes in plant phenology are influenced more by seasonal rainfall variation than by temperature (Eamus and Prior 2001; Lawrence 2005); seasonal changes display relatively high influences over reproductive factors (e.g., flowering and fruiting); and super-annual fluctuation cycles impact seedling recruitment (Bin et al. 2012; Kurten et al. 2018; Marod et al. 2004; Saikhammoon et al. 2023). Factors such as large water deficits are key triggers for leaf shedding, with peak shedding occurring during the dry season (Marod et al. 2023; Murphy and Lugo 1995; Zhang et al. 2014). Moreover, the ecological characteristics, structure, and dynamics of TDFs differ significantly from those of tropical rainforests (Gerhard and Hytteborn 1992). Therefore, understanding the dynamics of TDF communities is crucial for predicting trends in their ecological characteristics and for managing the stability of TDFs generally.

Long-term monitoring of tree survival and mortality using large, permanent plots provides valuable insights into TDF forest dynamics (Bunyavejchewin et al. 2019; Feeley et al. 2011; Ferreira et al. 2025). Variation in the survival and mortality of juvenile trees is a key factor in understanding demographic processes and community dynamics (Deb and Sundriyal 2008; McLaren et al. 2011; Norden et al. 2009), and plays significant roles in shaping the structure of old-growth forests and, by extension, their role as crucial habitats for diverse flora and fauna (Lombardi et al. 2011; Ma et al. 2014). Studying tree mortality processes can be challenging, as trees tend to have long lifespans and often exhibit low rates of mortality (Condit et al. 1999; Das et al. 2008) mediated by both intrinsic (e.g., tree size) and extrinsic (e.g., local biotic interactions and abiotic conditions) factors. The probability of tree mortality generally decreases as trees grow and develop greater resilience to environmental stress (Marod et al. 1999). Meanwhile, juvenile tree survival is determined by the physical and biotic attributes of the habitat as well as the physiological traits of the species (Aiba and Nakashizuka 2007). Compared to their older counterparts, juvenile trees compete less effectively for resources like water, nutrients, and light. Additionally, juvenile trees are often more influenced by abiotic factors (e.g., topography), since larger trees typically occupy more favorable habitats due to environmental filtering that occurs during early life stages (Phumphuang et al. 2024). Consequently, tree mortality and survival result from complex interactions among a range of limiting factors.

Additionally, species with various life-history strategies can be influenced differently by both intrinsic and extrinsic factors (Comita et al. 2009; Hubbell et al. 2001; Ma et al. 2014). A demographic trade-off between rates of growth vs. mortality is perhaps the best-established axis of life-history variation among tree species in both closed- and open-canopy forests (Wright et al. 2010). Shade tolerance and trade-offs between fast- and slow-growing species are key drivers of tree recruitment in tropical forest ecosystems (Rüger et al. 2020). Life-history strategies related to growth–mortality trade-offs reflect both tolerance to abiotic stressors (e.g., shade) and resilience to negative biotic interactions such as the density of conspecifics (Asanok et al. 2013). Similarly, seedling mortality may be influenced by density-dependent factors such as predators, herbivores, and pathogens (Zhu et al. 2018). The effects of conspecific density depend on species and local conditions (Hyatt et al. 2003; Asanok et al. 2013). For example, shade-tolerant species may coexist through conservative traits (e.g., thicker leaves) and persist in low-light environments due to their ability to withstand light limitation (Conti and Díaz 2013; Fine et al. 2006; Kitajima 1994; Kitajima and Poorter 2010). In contrast, light-demanding species may coexist through acquisitive traits (e.g., high specific leaf area), growing rapidly and opportunistically in response to light availability despite their sensitivity to shading from neighboring vegetation (Myers and Kitajima 2007; Zhu et al. 2018). Thus, light-demanding species are characterized by rapid growth, whereas shade-tolerant species exhibit higher survival rates (Li et al. 2023; McLaren and McDonald 2003). Thus, differences in life-history strategies likely represent key traits that shape species-specific responses to local abiotic and biotic conditions (Ma et al. 2014; McCarthy-Neumann and Kobe 2008).

In Thailand, TDFs comprise mixed deciduous forest and deciduous dipterocarp forest (Bunyavejchewin et al. 2011; Marod et al. 1999; Sakurai et al. 1998). These forest types have wildfire and drought stress as limiting factors (Marod et al. 2004; Wanthongchai et al. 2008), as well as long histories of anthropogenic disturbance (Delang 2002). Under these stressful conditions, TDF plant communities have evolved functional traits that increase their stability (Marod et al. 2004; Asanok et al. 2022). Recently, numerous studies have examined the effects of both biotic and abiotic factors on tree survival in tropical and subtropical regions (Comita et al. 2009; Hubbell et al. 2001; Ma et al. 2014), yet few have focused on TDFs, particularly those in Southeast Asia. In the present study, to explore local variation in tree survival in western

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Thailand, we used 30 years of tree dynamics' data from a 4 ha permanent plot in a mixed deciduous tropical dry forest. Our study investigated both intrinsic (tree size) and extrinsic (topography and local biotic neighborhood) factors and their impacts on tree survival. Specifically, we examined whether tree size plays a central role in determining tree survival, whether abiotic factors are more important drivers of tree survival than local biotic interactions, and whether the life-history strategy or ecological guild of a tree species has a greater influence on its survival. Our findings contribute to a deeper understanding of forest dynamics and offer valuable insights for the management of TDFs in Thailand and beyond.

Materials and methods

Study site

Our study took place in a mixed deciduous forest at Mae Klong Watershed Research Station, Thong Pha Phum District (14°30′–14°45′N, 98°45′–99°E), in western Thailand (Fig. 1a). The climate is subject to monsoons and annual rainfall normally exceeds 1650 mm, most of which falls during the rainy season from April to October. The mean temperature is about 27.5 °C, with a maximum of 39.1 °C in April and a minimum of 14.6 °C in December. The annual temperature showed a decreasing trend over 1993–2021. However, evaporation exhibited an increasing trend, contributing to drought conditions, particularly during the critical El Niño events of 1993–1995 and 1997–1998 (Marod et al. 2023). The study site is 150–350 m above sea level. The soils at the study site are classified as Alfisols, derived from sedimentary rock, gneiss, and limestone. The topsoil is ~ 20 cm deep with a sandy clay or sandy clay–loam texture and the subsoil, at a depth of ~ 2 m, consists of clay. Water drainage is moderate (Takahashi et al. 2011).

The prevailing forest type at the Mae Klong Watershed Research Station is developed mixed deciduous forest, nearly 30 m tall, with patches of deciduous dipterocarp forest on mountain ridges and dry evergreen forest along creeks. Normalized difference vegetation index (NDVI) shows a higher value in the northeastern than the southwestern region (Fig. 1b). The dominant tree species are mostly deciduous species such as *Pentacme siamensis*, *Xylia xylocarpa* var. *kerrii*, and *Pterocarpus macrocarpus* with the evergreen *Dipterocarpus alatus* along valleys (Marod et al. 1999; Saikhammoon et al. 2023). Within the mixed deciduous forest, undergrowing bamboos such as *Cephalostachyum pergracile* and *Bambusa tulda* dominate the middle layer, with heights reaching 10–15 m, and fires occur frequently in the dry (summer) season (Marod et al. 2002).

Data collection

In 1992, a 4 ha (200 m × 200 m) permanent plot was established on the south-facing slope of a mountain at the Mae Klong Watershed Research Station (Fig. 1c). All free-standing trees with a diameter at breast height (DBH; 1.30 m above ground) of at least 5 cm were tagged, measured, and identified to the species level. The plot was divided into 20 m × 20 m subplots, and the geographical coordinates of each tree within these subplots were recorded. Unidentified tree species were collected for identification and compared with specimens housed at the Forest Herbarium of the Department of National

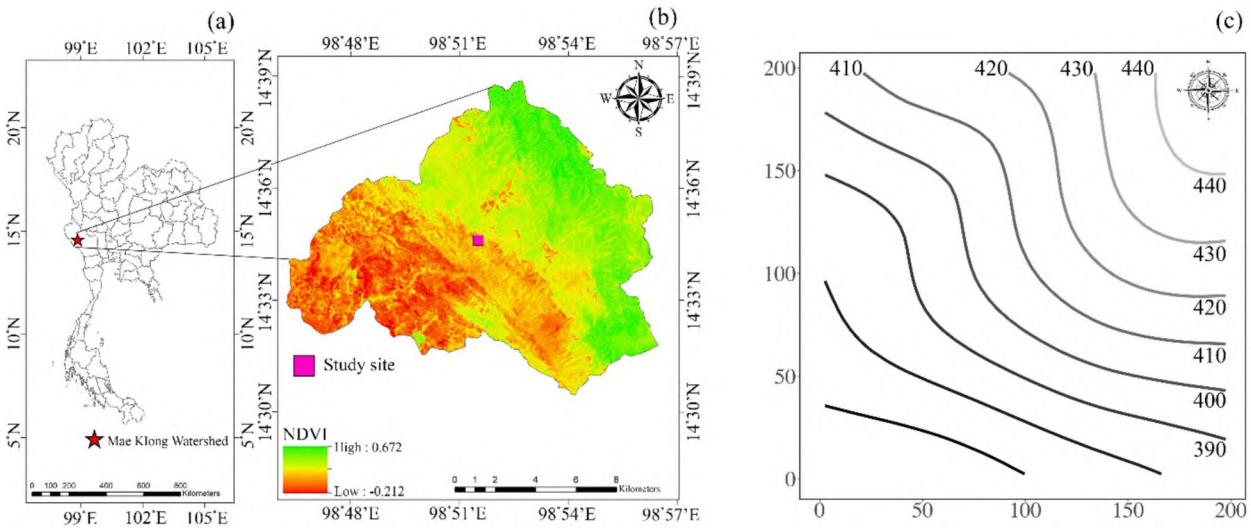


Fig. 1 The study area, located at Mae Klong Watershed Research Station, western Thailand (**a**); Normalized difference vegetation index (NDVI) value (**b**); contour map of the 4 ha permanent plot, showing the south-facing slope of a mountain at the Mae Klong Watershed Research Station (**c**)

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Parks, Wildlife and Plant Conservation. Tree censuses and measurements were conducted every 2 years in December from 1994 to 2022; additionally, trees ≥ 2 cm DBH were included since 1994. At each census, the mortality attributes of dead trees (standing dead, broken trunk, or uprooted) were also recorded. For our analysis, we used all census data from 1994 onward, including trees ≥ 2 cm in DBH.

Assessment of abiotic and biotic factors

Among biotic environmental factors, we analyzed only local factors related to plant community structure, i.e., the number of trees (reflecting tree density) and basal area (reflecting community dominance). Among abiotic factors, we analyzed only local topographic factors, i.e., elevation, slope, and convexity. However, we consider the omission of other biotic and abiotic factors, such as herbivory, pathogens, soil, and climate factors, a limitation of this study.

To assess the local-scale biotic neighborhood, we also recorded the total number of stems and the total basal area of all stems with a DBH ≥ 2 cm within each 20 m × 20 m subplot. The subplot dimensions were selected based on previous findings that tree species' interactions tend to dissipate beyond this range (Ma et al. 2014; Zhang et al. 2010). As abiotic factors, three topographic variables (mean elevation, slope, and convexity) were extracted from the field topographic surveys of each subplot. Mean elevation was calculated as the average of the four subplot corners. Slope was calculated as the average angle of the subplot relative to horizontal. Convexity was calculated as the mean elevation of each subplot minus the mean elevations of its eight neighboring quadrats, with the convexity of edge quadrats calculated as the elevation of the center point minus the mean elevation of the four corners. These variables were extracted using the *fgeo* package (Lepore et al. 2019) in R software (version 3.6.1).

Data analysis

To describe forest demography within the permanent plot, annual mortality and recruitment rates were calculated following the standard protocol proposed by Condit et al. (1999). Annual mortality was calculated using the formula $(\log N_1 - \log S_2)/t$, and annual recruitment was calculated as $(\log N_2 - \log S_2)/t$, where N_1 and N_2 represent the number of individuals alive in census 1 and census 2, respectively; S_2 is the number of individuals that survived to census 2; and t is the time interval between the two censuses, in years. These values were calculated using the *fgeo* package (Lepore et al. 2019) in R software (version 3.6.1).

We used generalized linear mixed models (GLMMs) to model the probability of individual tree survival over a 2-year interval as a function of initial tree size at the first census, along with abiotic and biotic neighborhood variables. The GLMMs applied in this study were logistic regressions, with tree survival as the binary response variable: 1 (alive) or 0 (dead) (Ma et al. 2014). Tree size (DBH) values were log-transformed before analysis to improve normality. All independent variables were standardized to have a mean of 0 and a variance of 1 using the "scale" function in R software (version 3.6.1). We assessed tree survival at three analytical scales. First, we performed a community-level analysis using all individuals within the permanent plot. Six fixed-effect variables were included: tree size (DBH), number of stems, total basal area, elevation, slope, and convexity, along with one offset term (census interval). Census date, species, and subplot were included as random-effect variables (Ma et al. 2014; Nutiprapun et al. 2023). Second, to assess the influence of species' life-history traits, we grouped species by ecological guild (i.e., pioneer or climax species) based on the Flora of Thailand database (https://botany.dnp.go.th/eflora/index.html) and conducted separate analyses for each category. Two shade-tolerance guilds, pioneer (light demanding) and climax species, were analyzed independently. Additionally, individuals were categorized into three growth stages based on DBH size class: saplings (2–5 cm DBH), poles (5–20 cm DBH), and mature trees (> 20 cm DBH). For both guild and stage analyses, the fixed- and random-effect variables remained consistent with the community-level model. Finally, we carried out a species-level analysis for the 20 most abundant species, based on the number of stems. All GLMMs were implemented using the *lme4* package (Bates et al. 2015) with the Laplace method (Bolker et al. 2009) in R software (version 3.6.1).

Results

Woody species composition

Over the period 1994–2022, in total 2864 individuals were monitored. As of 2022, the plot contained 109 species, representing 73 genera and 29 families. The tree density and basal area (≥ 2 cm DBH) were 238 individuals ha^{-1} and 17.60 $m^2\ ha^{-1}$, respectively. The diameter classes of all individuals (living and dead) showed a reverse-J-shaped distribution, indicating that some uneven-aged stands have been managed sustainably in the absence of major disturbance (Fig. 2). The most abundant species was *Croton mangelong*, followed by *Oroxylum indicum*, *Sterculia macrophylla*, *Millettia brandisiana*, and *Vitex canescens*.

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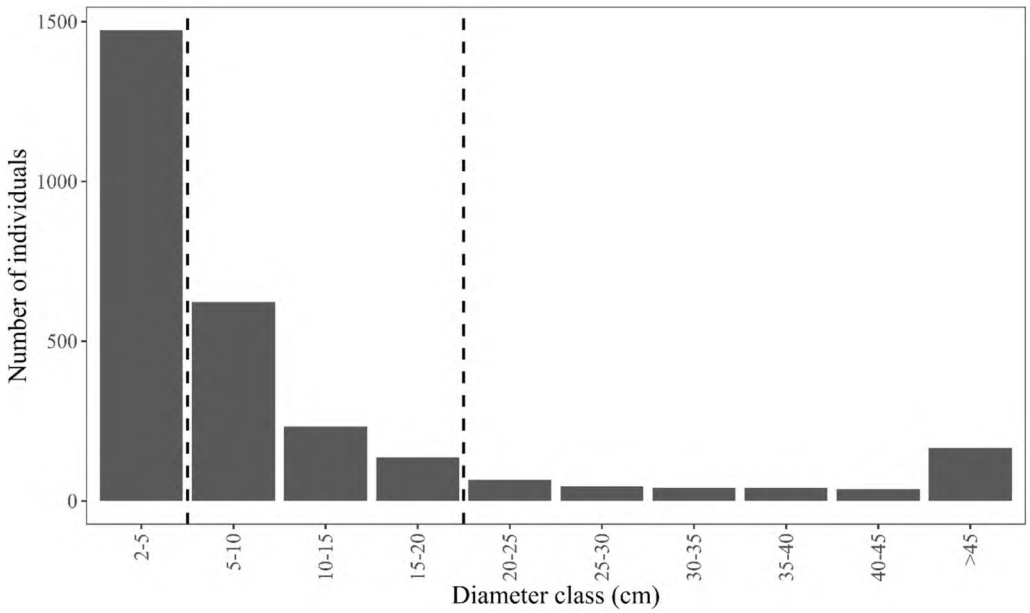


Fig. 2 Diameter class distribution of all individual trees ≥ 2 cm DBH in the 4 ha plot at Mae Klong Watershed Research Station, western Thailand. Dashed vertical lines indicating the separations between saplings (2–5 cm), pole sized (5–20 cm), and large trees (> 20 cm)

Survival dynamics

Annual mortality and recruitment rates differed slightly, at 3.87% year^{-1} and 3.06% year^{-1}, respectively (Fig. 3). Mortality rates did not differ among saplings, poles, and mature trees. Saplings had a high mortality rate during the first year, which decreased significantly over time ($p < 0.01$), whereas poles and large trees showed no significant differences in mortality throughout the study period, except for large trees in the current year (Fig. 4a). Among life-history guilds, pioneer species had a high mortality rate during the initial year and decreased significantly in the current year ($p < 0.01$), except for climax species, which showed a non-significant decrease (Fig. 4b).

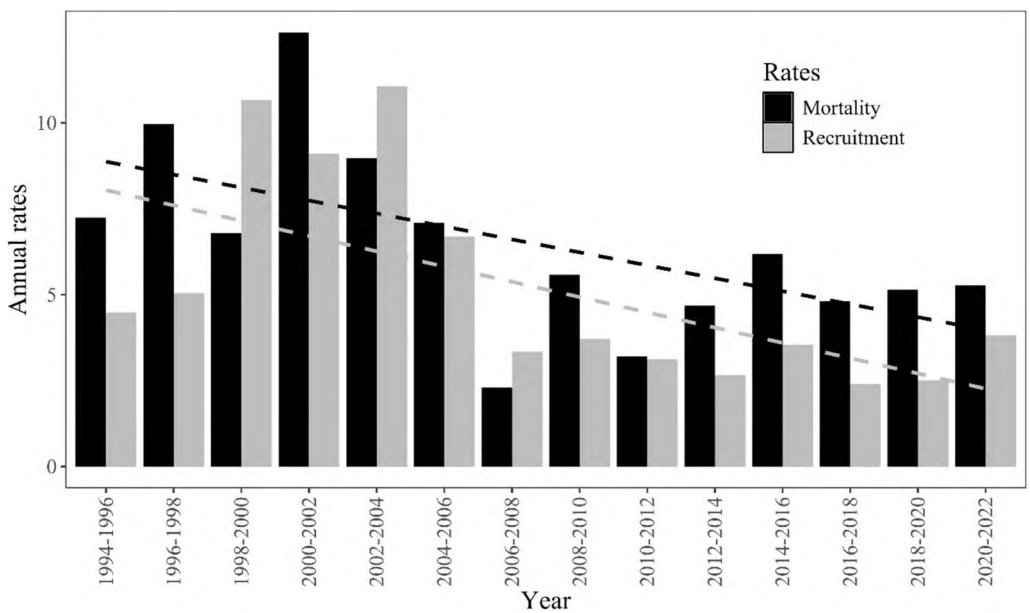


Fig. 3 Changes in tree mortality and recruitment rates over time (1994–2022) in the 4 ha permanent plot at Mae Klong Watershed Research Station, western Thailand. Annualized rates are shown as bar plots

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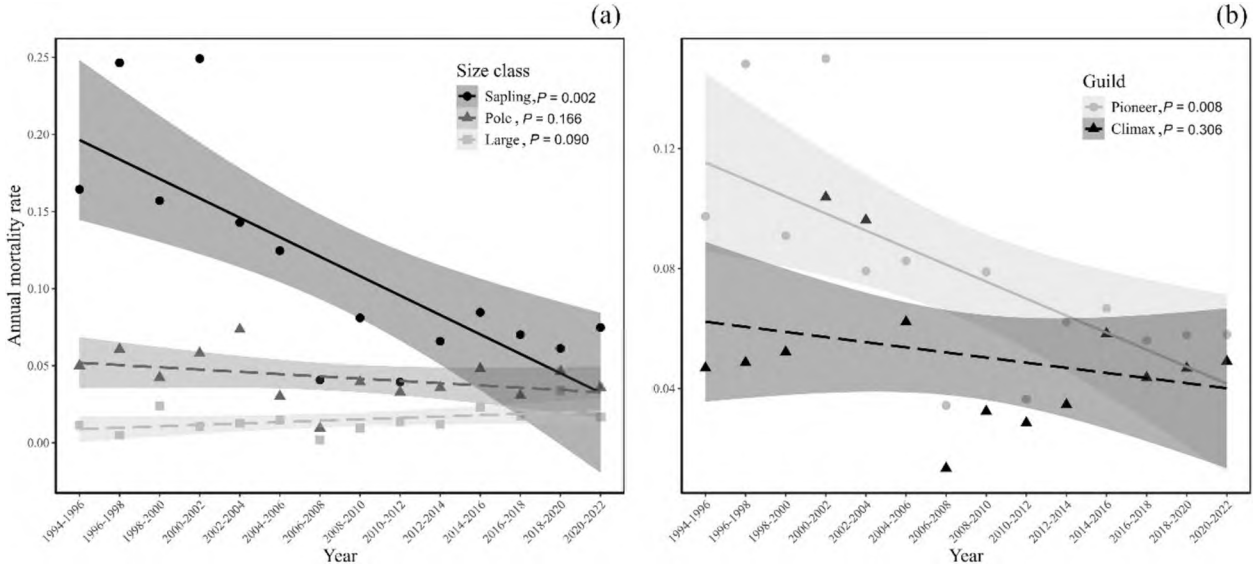


Fig. 4 Changes in mortality rate over time (1994–2022) in the 4 ha permanent plot at Mae Klong Watershed Research Station, western Thailand. Size class (**a**); ecological guild (**b**)

Table 1 Predictive and discriminative R^2 values in generalized linear mixed models of tree survival at the community, ecological guild, and size class levels

Tree class	Conditional R^2	Marginal R^2	ICC	RMSE	Abundance
Community					
All	0.655	0.248	0.542	0.373	2864
Ecological guild					
Climax	0.683	0.238	0.584	0.376	1362
Pioneer	0.680	0.248	0.575	0.357	1502
Size class					
Sapling	0.623	0.031	0.611	0.358	1481
Pole	0.723	0.074	0.701	0.369	986
Large	0.862	0.013	0.861	0.250	397

ICC, intraclass correlation coefficient; RMSE, root mean square error

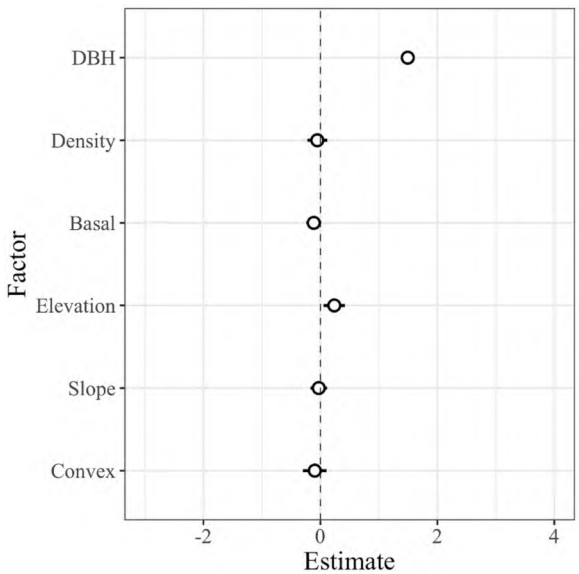


Fig. 5 Results of generalized linear mixed model analyses of the effects of tree size (DBH), biotic neighborhood (density and basal area), and local abiotic variables (elevation, slope, and convexity) on tree survival at the community level

Community-level analysis

The effects of tree size, biotic neighborhood (i.e., total tree density and basal area), and local abiotic variables (topography) on tree survival at the community level are shown in Table 1. Tree size had the strongest positive influence on tree mortality, with larger trees displaying a higher probability of survival. Elevation and total species basal area within subplots also affected tree survival, with elevation having a positive effect and basal area a negative effect (Fig. 5).

Guild-level analysis

We analyzed different life-history guilds to assess variations in the factors influencing tree survival. We found notable differences in the relative importance of these factors between pioneer and climax species. Across both pioneer and climax species, DBH was significantly positively associated with tree survival

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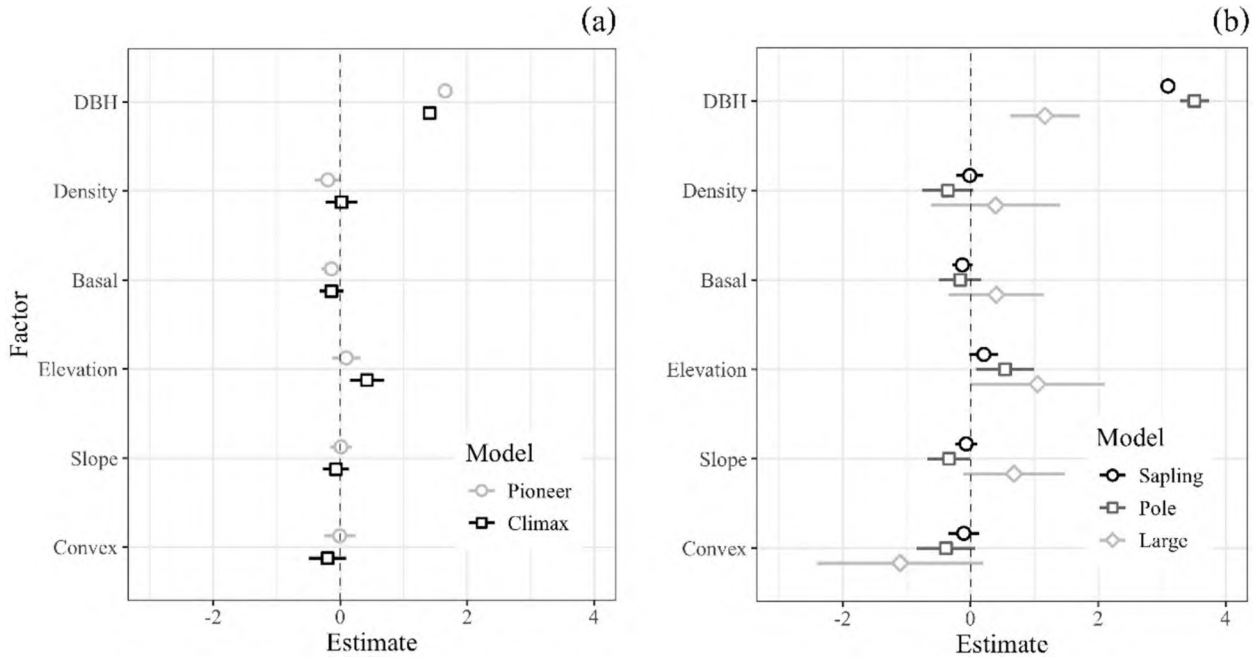


Fig. 6 Results of generalized linear mixed model analyses of the effects of tree size (DBH), biotic neighborhood (density and basal area), and local abiotic variables (elevation, slope, and convexity) on tree survival at the ecological guild (**a**) and size class (**b**) levels

(Fig. 6a). Reflecting the strong positive effect of DBH, survival was higher among large trees than among small ones. Tree density variables showed only negative relationships with survival for pioneer species, indicating increased mortality for clumped patterns of establishment. Meanwhile, elevation had a significant positive effect on the survival of climax species, indicating higher mortality in lowland climax species.

We also analyzed three growth stages (sapling, pole, and mature tree) separately and found that the factors influencing survival varied by size class (Fig. 6b). Tree size had a significant positive effect on survival for both the sapling and pole size classes ($p < 0.001$). In the pole group, survival decreased with higher tree density and basal area. Abiotic factors showed no significant correlation with survival in any of the size classes. Furthermore, none of the variables examined had significant associations with tree survival in the largest size class.

Species-level analysis

GLMMs were conducted on 20 species, and the effects of tree size, as well as abiotic and biotic factors, on tree survival varied (Table 2). Tree size (DBH) had a strongly positive influence on survival across all species ($p < 0.001$). Basal area had a significant positive effect for *Lagerstroemia floribunda* and *Colona floribunda* ($p < 0.05$), but a significant negative effect for *Xylia xylocarpa* ($p < 0.05$). Of the 20 species examined, mortality was influenced by abiotic factors in five species, but the effects varied in both extent and direction. Elevation had a significant positive effect only for *Sterculia macrophylla* ($p < 0.05$). Slope had a negative effect for *Oroxylum indicum* and *Senna garrettiana* ($p < 0.05$). Convexity was negatively associated with mortality in *Markhamia stipulata* ($p < 0.05$) but positively associated in *Lagerstroemia venusta* ($p < 0.01$).

Discussion

This study used plant community data from a 4 ha plot within a mixed deciduous tropical dry forest in western Thailand. Therefore, our results can reveal only trends in TDF dynamics and may not be representative of the entire TDF forest landscape. Furthermore, we selected only topographic factors to represent abiotic factors, and biotic factors included only DBH class and local factors such as tree density and basal area. Therefore, our findings cover a limited range of the environmental factors that affect plant community establishment, excluding factors such as soil properties, light conditions, and climate factors.

Effect of tree size

Tree size had the strongest effect on survival in all but the largest size class. Empirical studies on tree survival indicate that mortality rates tend to decrease as tree size increases (Marod et al. 1999), although there is no

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Table 2 Results of generalized linear mixed model analyses of the effects of tree size (DBH), biotic neighborhood (density and basal area), and local abiotic variables (elevation, slope, and convexity) on tree survival for 20 selected species

Species	(Int)	DBH	Density	Basal	Elevation	Slope	Convexity	Guild	Abundance
Croton mangelong	−3.263	**1.724*** **	−0.521	−0.290	−0.638	0.489	0.299	Pioneer	202
Oroxylum indicum	−5.891	**3.217*** **	−0.158	0.318	0.52	**−0.624***	−0.258	Pioneer	162
Sterculia macrophylla	−7.282	**2.929*** **	−0.257	0.027	**1.574***	0.397	0.041	Pioneer	135
Millettia brandisiana	−3.322	**1.970*** **	1.013	0.649	0.509	0.568	0.335	Climax	109
Vitex canescens	−6.503	**3.685*** **	0.609	0.055	0.388	-0.565	0.213	Climax	108
Senna garrettiana	−6.958	**3.542*** **	0.248	1.190	0.201	**−1.152***	0.303	Pioneer	101
Berrya cordifolia	−7.573	**2.663*** **	−0.389	−0.336	−0.275	0.465	0.002	Pioneer	86
Colona floribunda	−6.394	**2.783*** **	0.191	**1.116***	1.046	−0.627	0.454	Pioneer	80
Bauhinia malabarica	−9.938	**3.959*** **	0.829	0.515	0.004	0.863	0.349	Pioneer	77
Mitragyna rotundifolia	−4.047	**3.662*** **	−0.292	−2.321	0.848	0.779	0.159	Climax	67
Xylia xylocarpa	−7.078	**4.397*** **	0.235	**−1.411***	0.619	-0.694	−0.918	Climax	67
Markhamia stipulata	−6.032	**4.382*** **	−0.881	-0.973	1.555	−1.630	**−1.842***	Climax	62
Bombax anceps	−5.426	**2.864*** **	1.049	−0.223	-0.422	−0.581	−0.735	Pioneer	61
Terminalia nigrovenulosa	−5.100	**3.233*** **	−0.906	0.181	-0.259	−0.426	−1.278	Climax	61
Bauhinia viridescens	−11.547	**6.905*** **	0.449	2.796	1.380	0.183	−0.556	Pioneer	56
Lagerstroemia floribunda	−2.502	**1.777*** **	−0.873	**2.032***	0.209	-0.262	0.163	Climax	48
Lagerstroemia venusta	−7.474	**3.741*** **	1.129	−0.871	-1.025	-1.349	**3.431****	Climax	44
Wrightia arborea	−1.663	**3.233*** **	0.137	2.800	-1.095	-0.043	−0.184	Climax	44
Canarium subulatum	−6.391	**2.686*** **	−0.427	0.025	1.108	-0.130	−1.085	Climax	43
Garuga pinnata	−4.030	**1.888*** **	0.613	0.369	0.059	-0.065	0.069	Climax	42

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

consensus on the precise nature of the relationship. In this study, tree size consistently displayed the strongest positive influence on survival across nearly all species and categories. However, when trees were categorized into growth stages (size classes), the effect of size shifted from strongly positive in the sapling and pole classes to slightly negative in the large tree class (Fig. 6b). Juvenile tree survival is determined by physical and biotic environmental factors within the habitat, as well as species' physiological traits (Aiba and Nakashizuka 2007). This implies that while survival generally increases with tree size, mortality may begin to rise again among the largest individuals, which may be at greater risk of being uprooted by strong winds during the rainy season (Marod et al. 1999; Silvério et al. 2019). Another possible explanation is based in metabolic ecology theory, which proposes that different size classes receive and utilize similar amounts of energy (Ma et al. 2014). This idea is further supported by findings from tropical rainforests in the Americas (McDowell et al. 2018), where rising temperatures and prolonged droughts have been linked to increasing tree mortality, particularly among large trees. To that end, drought and fire disturbance also affect the dynamics and regeneration of TDFs (Gerhard and Hytteborn 1992; Marod et al. 2002; Yatar et al. 2024). In our study area, however, both mean and maximum temperature decreased slightly over the study period, despite the tendency toward drought during El Niño events (Marod et al. 2023). Fires frequently follow drought events, particularly in the dry season, and increase seedling and sapling mortality (Marod et al. 2002). Diseases are another potential cause of mortality in large trees, with old-growth tissues tending to be less resistant to disease (Chin et al. 2023; Das et al. 2016). Overall, our findings imply that while larger trees generally have higher rates of survival, their risk of mortality may eventually increase again due to environmental and physiological pressures (Ma et al. 2014; McDowell et al. 2018).

Effects of abiotic and biotic neighborhoods

In this study, abiotic factors (elevation, slope, and convexity) were incorporated into the GLMMs at the community level as well as for various guilds and/or species. Overall, elevation and convexity had positive effects on climax species and negative effects on pioneer species (Fig. 6a). At higher elevations, drought-resistant deciduous species had a competitive advantage and displayed higher rates of survival; however, on exposed ridges with high convexity, their survival was lower. Takahashi et al. (2019) reported that soil moisture content was lower at higher elevations, leading to severe drought and increased mortality on ridges. Other potentially

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important variables, such as soil properties and light availability, were not included in our analysis and may have influenced our results. However, it is notable that our study was restricted to a 4 ha plot. The influence of abiotic factors on tree survival may vary across limited spatial scales and might be more prominent in studies conducted at broader scales.

Among the local biotic factors examined, tree density and total basal area had significant negative effects on tree survival for some guilds (pioneer and climax species) and species. These findings are consistent with many previous studies in tropical forests (Hubbell et al. 2001; Marod et al. 1999) and support the concept of negative density- and frequency-dependent effects (Hubbell et al. 2001; Freckleton and Lewis 2006). These mechanisms are widely recognized as important drivers of tree population dynamics and may play significant roles in regulating local species diversity (Hubbell et al. 2001; Ma et al. 2014). Our study also revealed positive effects of biotic neighborhood variables on the survival of some species, in contrast to the patterns suggested by earlier studies. Ma et al. (2014) proposed that such inconsistencies may be explained by the complexity of habitat structure and the spatial distribution of trees. Clumping appeared to be relatively low in our 4 ha plot, possibly due to the dominance of bamboo species in the sub-canopy layer (Marod et al. 1999), which may have altered competitive dynamics and neighborhood effects. Several previous studies have suggested that bamboo in the sub-canopy layer is an important factor that affects tree mortality, recruitment, and regeneration in natural tropical forests (Larpkern et al. 2011; Griscom and Ashton 2003; Guilherme et al. 2004).

Variation among ecological guilds and species

The factors examined in this study had various impacts on the survival of different ecological guilds. For example, the survival of pioneer (i.e., light-demanding) and climax (i.e., shade-tolerant) species was positively associated with DBH. Larger stems develop stronger trunks that show greater resistance to environmental stresses (Sestras et al. 2025). A previous study found that differences in tree growth rates explained the large variation in drought resistance and recovery among geographic regions (Tao et al. 2024). In pioneer species, survival was negatively associated with tree density. Similarly, a previous study reported that the mortality of pioneer species (e.g., *Colona floribunda*, *Sterculia macrophylla*, and *Berrya ammonilla*) generally increased with increasing density due to competition for available resources (Marod et al. 1999), which is consistent with Janzen–Connell effects (Clark and Clark 1984). Tree density likely affects juvenile mortality because predators, herbivores, and pathogens often exhibit density-dependent behavior (Clark and Clark 1984; Freckleton and Lewis 2006). Tree-density effects may depend on the species and on local conditions (Hyatt et al. 2003). For example, light interception by the foliage and branches in a dense forest canopy profoundly affects the growth of pioneer species (Matsuo et al. 2021; Beauchamp et al. 2025). In climax species, survival is positively influenced by elevation. Generally, the survival of mixed deciduous forest climax species is related to topographic variables such as elevation, and is positively associated with abundant light and drought (Takahashi et al. 2019). However, their establishment may be hampered by fire, which often arises along ridges and consumes the aboveground portions of pioneer species. Such fires create large gaps that allow more light to penetrate, which in turn facilitates liana colonization and, counterintuitively, may rapidly lead to even denser shading of the understory (Addo-Fordjour et al. 2020). This process may explain why the survival of climax and late-successional species was more positively associated with elevation in our 4 ha plot (Comini et al. 2023).

All of the variables analyzed in this study influenced the survival of at least some of the 20 species examined. We expected species within the same guild to respond similarly to their environments (Ma et al. 2014), and found a positive relationship with DBH in all species at the ecological guild level. Tree size is positively associated with basal area, and we found that both pioneer species (e.g., *Colona floribunda*) and climax species (e.g., *Lagerstroemia floribunda*) exhibited positive relationships between survival and basal area. Thus, stem size is highly important for tree survival in TDFs. However, when the influence of other local abiotic and biotic variables on tree species survival was considered, we found no association at the ecological guild level. Rather, we detected no relationship between pioneer species survival and tree density, and no relationship between climax species survival and elevation. This contradiction between the guild- and species-level analysis results may be explained by the fact that species within particular guilds often differ in other characteristics that influence survival patterns (Ma et al. 2014; Wang et al. 2009), which is consistent with the considerable interspecific variation within guilds reported in previous studies on plant survival in tropical and subtropical forests (Comita et al. 2009; Ma et al. 2014). For example, our results show that survival of the pioneer species *Sterculia macrophylla* was positively associated with elevation; however, in both pioneer species examined in this study (*Oroxylum indicum* and *Senna garrettiana*), survival was negatively associated with slope. For climax species such as *Lagerstroemia venusta* and *Markhamia stipulata*, survival was associated with convexity. These results imply that the survival of woody

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species in TDFs is associated with diverse environmental factors, likely because species possess different functional traits despite their coexistence within the same ecological communities (Paine et al. 2011; Wright et al. 2010). Our results imply that topographic heterogeneity promotes diversity in TDFs at both the species level via environmental filtering (i.e., local abiotic and biotic factors) and at the population level via functional responses to the density of neighboring vegetation (O'Brien and Escudero 2022).

Conclusion

Our findings imply that tree size, biotic interactions, and niche partitioning related to topography all contribute to the regulation of the TDFs studied in western Thailand. Although this study had limitations, in that we did not analyze all possible environmental factors influencing tree survival, our results imply that tree survival did not increase consistently with size, and varied substantially among species in the study area. Moreover, the relative importance of the examined factors differed across ecological guilds and species. Among abiotic factors, only elevation had a significant positive effect on survival in shade-tolerant species. Of the biotic factors examined, tree density and total basal area significantly impacted tree survival in certain guilds and species. These findings imply that, although the implications for species diversity require further investigation, efforts to understand, conserve, and manage TDFs should carefully consider both intrinsic (e.g., tree size and life-history traits) and extrinsic (e.g., topography and neighborhood effects) drivers of forest dynamics. A deeper understanding of both intrinsic and extrinsic effects on forest dynamics would accelerate successful forest restoration. Moreover, forest conservation and management policies should take a wide range of variables into account to have the best chance of improving outcomes.

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Author contributions
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Data availability
The data used during the current study is available on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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